ASSIGNMENT-4: LISTS

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1. Find the sum of all elements of a list.

In [1]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
s=sum(l)
print('Sum of all elemnets in the list: ',s)
```

```
Enter the number of elements: 3
Enter value 1 : 8
Enter value 2 : 6
Enter value 3 : 4
Sum of all elemnets in the list:  18
```

2. Find the maximum and minimum element of a list.

In [2]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
maxm=max(l)
minm=min(l)
print(f'Maximum element: {maxm} \nMinimum element: {minm}')
```

```
Enter the number of elements: 4
Enter value 1 : 4
Enter value 2 : 8
Enter value 3 : 5
Enter value 4 : 12
Maximum element: 12
Minimum element: 4
```

3. Find the position of maximum and minimum element in a list.

In [3]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
maxm=l.index(max(l))
minm=l.index(min(l))
print(f'Maximum element index: {maxm} \nMinimum element index: {minm}')
```

```
Enter the number of elements: 4
Enter value 1 : 45
Enter value 2 : 87
Enter value 3 : 5
Enter value 4 : 98
Maximum element index: 3
Minimum element index: 2
```

4. Print all the elements in a list that are greater than a given value.

In [4]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
```

```
    l.append(int(input(f"Enter value {i+1} : ")))
l_greater=[]
n=int(input('Enter value to compare: '))
for i in l:
    if i>n:
        l_greater.append(i)
print(f'Elements greater than {n} are: ', l_greater)
```

```
Enter the number of elements: 4
Enter value 1 : 1
Enter value 2 : 8
Enter value 3 : 9
Enter value 4 : 2
Enter value to compare: 5
Elements greater than 5 are:  [8, 9]
```

5. Find the average of all elements present in a list.

In [5]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
avg=sum(l)/len(l)
print('Average of all elements in the list: ',avg)
```

```
Enter the number of elements: 4
Enter value 1 : 5
Enter value 2 : 9
Enter value 3 : 1
Enter value 4 : 15
Average of all elements in the list:  7.5
```

6. Take two lists and print only their common elements.

In [6]:
```
l1, l2 = [], []
for i in range(int(input("Enter the number of elements for list 1: ")):
    l1.append(int(input(f"Enter value {i+1} : ")))
for i in range(int(input("Enter the number of elements for list 2: ")):
    l2.append(int(input(f"Enter value {i+1} : ")))
print('Common elements are: ')
for i in l1:
    if i in l2:
        print(i, end=' ')
```

```
Enter the number of elements for list 1: 5
Enter value 1 : 4
Enter value 2 : 8
Enter value 3 : 2
Enter value 4 : 8
Enter value 5 : 7
Enter the number of elements for list 2: 3
Enter value 1 : 8
Enter value 2 : 7
Enter value 3 : 4
Common elements are:
4 8 8 7
```

7. Check whether the two lists are equal or not.

In [7]:
```
l1, l2 = [], []
for i in range(int(input("Enter the number of elements for list 1: ")):
    l1.append(int(input(f"Enter value {i+1} : ")))
for i in range(int(input("Enter the number of elements for list 2: ")):
    l2.append(int(input(f"Enter value {i+1} : ")))
if l1==l2:
    print('Lists are equal.')
```

```
else:
    print('Lists are not equal.')
```

```
Enter the number of elements for list 1: 3
Enter value 1 : 5
Enter value 2 : 9
Enter value 3 : 4
Enter the number of elements for list 2: 3
Enter value 1 : 5
Enter value 2 : 9
Enter value 3 : 4
Lists are equal.
```

8. Take a list and reverse its elements using slicing.

In [8]:
```
l = []
for i in range(int(input("Enter the number of elements: "))):
    l.append(input(f"Enter value {i+1} : "))
l=l[::-1]
print('Reversed List: ', l)
```

```
Enter the number of elements: 4
Enter value 1 : Srabani
Enter value 2 : 5.97
Enter value 3 : :"\
Enter value 4 : Hello World
Reversed List:  ['Hello World', ':"\\', '5.97', 'Srabani']
```

9. Print only those numbers present in a list which are divisible of 5.

In [9]:
```
l = []
for i in range(int(input("Enter the number of elements: "))):
    l.append(int(input(f"Enter value {i+1} : ")))
for i in l:
    if i%5==0:
        print(i, end=' ')
```

```
Enter the number of elements: 5
Enter value 1 : 4
Enter value 2 : 15
Enter value 3 : 35
Enter value 4 : 4
Enter value 5 : 8
15 35
```

10. Given a list of numbers, return 'True' if the first and last numbers are same.

In [10]:
```
l = []
for i in range(int(input("Enter the number of elements: "))):
    l.append(int(input(f"Enter value {i+1} : ")))
if l[0]==l[-1]:
    print('True')
else:
    print('False')
```

```
Enter the number of elements: 3
Enter value 1 : 5
Enter value 2 : 4
Enter value 3 : 5
True
```

11. Find the second largest and second smallest element from a list

In [12]:
```
l = []
for i in range(int(input("Enter the number of elements: "))):
```

```
    l.append(int(input(f"Enter value {i+1} : ")))
l.sort()
lar2, sm2 =l[-2], l[1]
print(f'2nd Largest: {lar2}\n2nd Smallest: {sm2}', )
```

```
Enter the number of elements: 5
Enter value 1 : 1
Enter value 2 : 8
Enter value 3 : 7
Enter value 4 : 6
Enter value 5 : 3
2nd Largest: 7
2nd Smallest: 3
```

12. Search an element present in a list using linear search.

In [13]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
n=int(input('Enter element to search: '))
for i in l:
    if i==n:
        print(f'{i} found at index {l.index(i)}')
        break
else:
    print(f'{n} not found!!!')
```

```
Enter the number of elements: 4
Enter value 1 : 7
Enter value 2 : 9
Enter value 3 : 1
Enter value 4 : 6
Enter element to search: 0
0 not found!!!
```

13. Search an element of a list using binary search.

In [14]:
```
l = []
for i in range(int(input("Enter the number of elements: ")):
    l.append(int(input(f"Enter value {i+1} : ")))
n=int(input('Enter element to search: '))
l.sort()
low,high=0,len(l)-1
while(low<=high):
    mid=(low+high)//2
    if n==l[mid]:
        print(f'{n} found at index {l.index(n)}')
        break
    elif (n<l[mid]):
        high=mid-1
    else:
        low=mid+1
else:
    print(f'{n} not found!!!')
```

```
Enter the number of elements: 5
Enter value 1 : 1
Enter value 2 : 8
Enter value 3 : 4
Enter value 4 : 5
Enter value 5 : 3
Enter element to search: 5
5 found at index 3
```

14. Generate random numbers within a given range and store them in a list.

In [15]:
```
import random
l= []
n=int(input('Enter length of list: '))
start=int(input('Enter start value: '))
stop=int(input('Enter stop value: '))
for i in range(n):
    l.append(random.randint(start, stop))
print(l)
```

```
Enter length of list: 5
Enter start value: 10
Enter stop value: 30
[26, 16, 14, 27, 29]
```

15. Split the even and odd elements of a list into two different lists.

In [16]:
```
l = []
for i in range(int(input("Enter the number of elements: "))):
    l.append(int(input(f"Enter value {i+1} : ")))
l_even, l_odd = [],[]
for i in l:
    if i%2==0:
        l_even.append(i)
    else:
        l_odd.append(i)
print('Odd List: ', l_odd)
print('Even List: ', l_even)
```

```
Enter the number of elements: 5
Enter value 1 : 465
Enter value 2 : 84
Enter value 3 : 15
Enter value 4 : 48
Enter value 5 : 15
Odd List:  [465, 15, 15]
Even List:  [84, 48]
```

16. Add two matrices

In [17]:
```
a = [[1,2,3],
    [4,5,6]]
b = [[1,3,5],
    [2,1,3]]
c = [[0,0,0],
    [0,0,0]]
for i in range(len(a)):
    for j in range(len(a[i])):
        c[i][j] = a[i][j] + b[i][j]
print(c)
```

```
[[2, 5, 8], [6, 6, 9]]
```

17. List Comprehension

In [18]:
```
l1 = [i for i in range(2,21,2)]
l2 = [i**2 for i in range(0,11)]
l3 = [i[0] for i in 'Hello, Srabani here...'.split()]
print("Even numbers within 20 : ", l1)
print("Squares from 0 to 10 : ", l2)
print("First letter of each word : ", l3)
```

```
Even numbers within 20 :  [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]
Squares from 0 to 10 :  [0, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
First letter of each word :  ['H', 'S', 'h']
```